UNDERSTANDING HOW POST-TRANSLATIONAL ACETYLATION AFFECTS THE FUNCTION OF RNA BINDING PROTEINS

by

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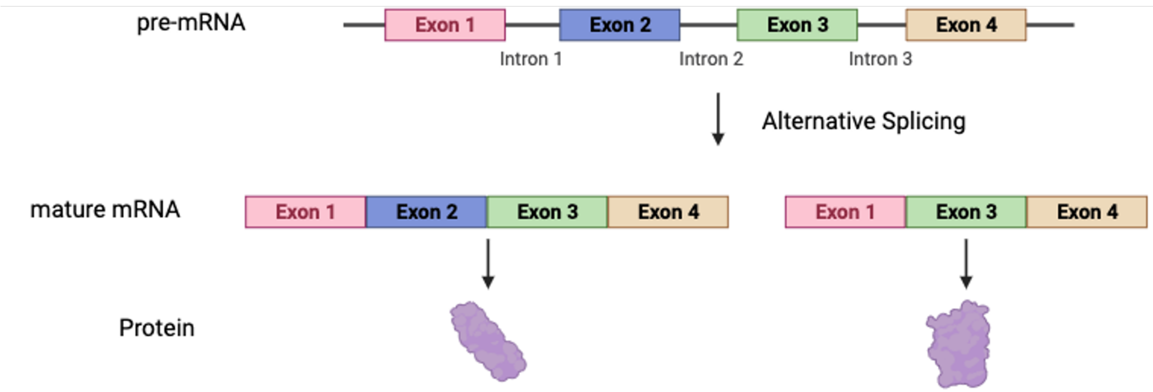
Abstract

This project aims to understand how post-translational acetylation affects the function of RNA binding proteins. Polypyrimidine tract binding protein 1 (PTBP1) is the RNA binding protein (RBP) that will be used in this experiment. This RBP is made of four RNA binding domains that are joined by three linker regions. PTBP1 regulates alternative splicing, which is the process of either including or excluding a regulated exon when forming a mature mRNA. Mis-regulation of alternative splicing can lead to neurodegenerative diseases because PTBP1 acts as a repressor for exons that are necessary for neuronal development. When a lysine residue is acetylated, it is left with a neutral charge. Lysine can hydrogen bond to the RNA prior to acetylation due to their opposite charges. Thus, we hypothesize that lysine acetylation can alter PTBP1 RNA binding properties. To test this, a lysine to glutamine mutant is created, specifically K424Q. We created a mutant gene using two-step PCR and cloned into mammalian expression vector pcDNA3.1+. We will assay the mutant for protein expression and splicing activity using wild type PTBP1 as a control.

Introduction

The central dogma of biology explains that DNA undergoes transcription, which is where the primary transcript is spliced and processed by adding a 5'cap and a poly-A tail. This process allows it to reach the cytoplasm, where translation occurs. Translation is where processed mRNA uses ATP and tRNA to link mRNA codons with proteins that are being built by amino acids. In summary, the central dogma goes from DNA, pre-mRNA, RNA, and to a protein. This research focusses on a post-translational process, so it occurs during the transition from an RNA to a protein.

This lab uses Polypyrimidine Tract Binding Protein 1 (PTBP1) to determine how post-translational acetylation affects the function of RNA Binding proteins (RBP). PTBP1 is an RBP that is made of 4 RNA binding domains that are connected by 3 linker regions. This type of protein is found in neuronal progenitor cells, which is why this research has implications to Alzheimer's and Parkinson's disease. RNA Binding proteins, such as PTBP1, regulate a process known as alternative splicing. Alternative splicing is the process of either including or excluding a constitutive exon when forming a mature mRNA. This process begins with a pre-mRNA that is a chain of alternating exons and introns. Exons are coding regions and introns are non-coding regions. Due to this, the introns are cut out by the spliceosome when forming a mature mRNA. The mature mRNA consists of only exons, which are again the coding region. A singular pre-mRNA can result in many mature mRNA's that code for different proteins. Below is an image representing this described process.

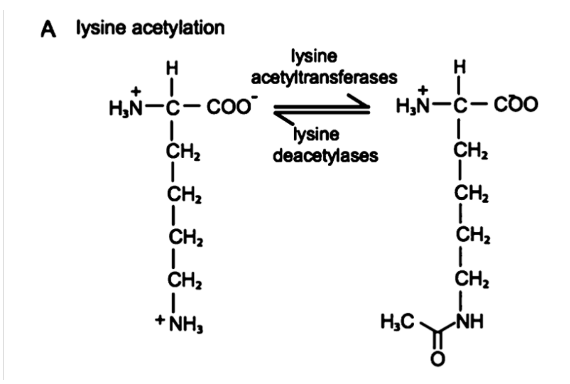


(1)

In figure 1, the blue exon 2 is the constitutive exon, or regulated exon. This exon can be seen in the code for protein 1 on the left but is absent in protein 2 on the right. A singular pre-mRNA can be spliced in many ways, resulting in proteins that are related but have different functions. Therefore, this process results in an increase in protein diversity.

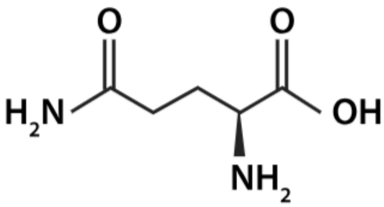
As revealed above, PTBP1 is used to regulate alternative splicing. When there is mis regulation of alternative splicing, neuronal diseases such as Parkinson's and Alzheimer's disease can develop. RBPs regulate alternative splicing by binding to cis regulatory elements of pre-mRNA substrates and can either enhance or disrupt the formation of a functional spliceosome. There are Serine-Arginine (SR) proteins that recruit and stabilize interactions by bridging the binding domain to the pre-mRNA transcript. SR proteins are splicing enhancers. There are also Heterogeneous ribonucleoprotein particles (hnRNPs) that bind to exonic motifs and acts as a suppressor. This protein blocks access of the spliceosome to the polypyrimidine tract. From this, it is revealed that SR proteins and hnRNPs have opposite effects of regulation. PTBP1 is a type of hnRNP, so the RBP that is focused on in this lab is a suppressor.

Mis regulation of alternative splicing can lead to neurodegenerative diseases because PTBP1acts as a repressor for exons that are necessary for neuronal development. When a lysine residue is acetylated, it is left with a neutral charge. A normal lysine residue has a positive charge and can hydrogen bond to the RNA prior to acetylation due to their opposite charges. The RNA tract is known to have a negative charge, which is why the lysine residue can no longer bind to the RNA after acetylation. A neutrally charged lysine is not expected to have any binding affinity to a negatively charged RNA tract. The specific PTBP1 mutation that is focused on throughout this experiment is K424Q, or lysine 424 glutamine. This lysine to glutamine mutant is created to mimic an acetylated lysine because the side chain of a glutamine amino acid resembles an acetylated lysine. In (2), the mechanism for lysine acetylation is shown, carried out by the enzyme lysine acetyltransferases in the forward direction.



(2)

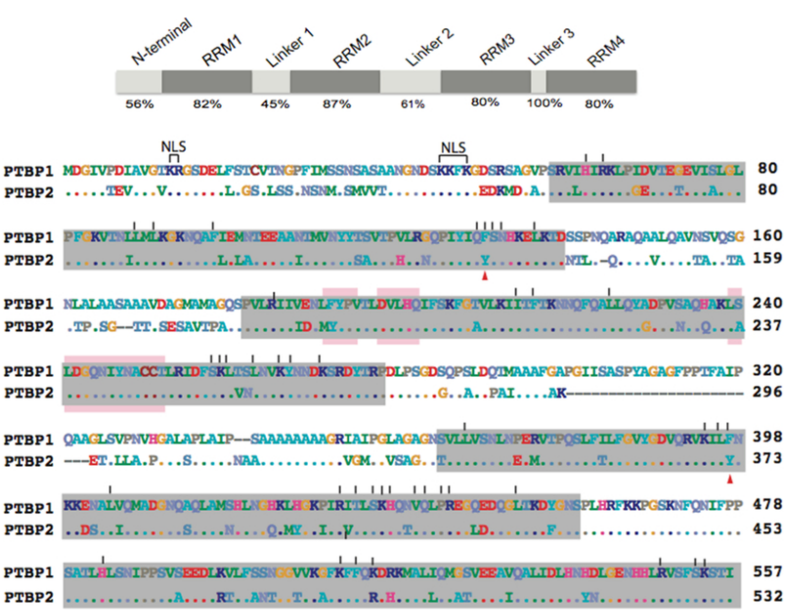
This image shows the lysine losing its positive charge when an acetyl group is added in its place. Shown below in (3) is the amino acid glutamine that can be used to resemble the acetylated lysine residue on the right of (2).



(3)

Since the amino acid glutamine has the same characteristics as an acetylated lysine, this mutation from lysine to glutamine is made to mimic the acetylated lysine. It is not possible to directly perform lysine acetylation, so generating this lysine to glutamine mutant will help indicate if lysine acetylation affects the function of the RNA binding protein.

The RBP of interest in this lab is PTBP1 and the full amino acid sequence for this is shown in (4). The 4 RBPs that make up the PTBP1 are recognition motif (RRM) 1, RRM2, RRM3, and RRM4. The K424 that is generated during this experiment is found on RRM3, with a percent acetylation of 11.70 ± 6.20. This Lysine acetylation occurs under splicing conditions with Hela cell nuclear extract. It is also known that the most acetylated lysine residues occur on RRM3, where K424 is found. The K424 is where the lysine to glutamine mutation is generated on the full length PTBP1 strand of 557 residues.



(4)

For this specific experiment, only the top strand of each row labeled PTBP1 is regarded. Below in (5) is the location of the K424 residue when looking at the 3D structure of PTBP1.


(5)

This illustrates RBD 3 and 4 of the PTBP1. The aromatic shape on the far right is the CUCUCU RNA hexamer and the K424 is unique because it forms a salt bridge with this hexamer.

Experimental

To begin this research, a 1st polymerase chain reaction (PCR) was completed by generating a forward and reverse fragment with the K424Q primer. The following reagents were added into two PCR tubes labeled “K424 Forward” and “K424 Reverse:” 34.5 μL of nH_2O, 10 μL of 5x Phusion Buffer, 1.0 μL of DMSO, 1.0 μL of 10 mM dNTPs, 1.0 μL of DNA template, and 0.5 μL of Phusion Plus DNA polymerase. In the forward PCR tube, 1.0 μL of Forward Mutant Primer and 1.0 μL of EcoRV Reverse Primer were added. In the reverse PCR tube, 1.0 μL of Reverse Mutant Primer and 1.0 μL of BamH1 Forward Primer were added. These reagents were resuspended to mix after being added to the respective PCR tube. The DNA polymerase was the final reagent added and only taken out of the -20 °C freezer at the time of use. These two mixtures were run in the thermocycler under conditions for Phusion Plus DNA Polymerase. Each PCR product was checked on a 1% agarose gel by loading 5 μL of Generuler 1 kb DNA ladder into the first lane on the gel. For each sample, 3 μL of the PCR product was mixed with 2 μL of

DNA loading dye. These two mixtures were injected into separate lanes on the gel and ran for 30 minutes at 90 V. The gel was then imaged on the OmegaLum at 302 nm. The same process was completed on a 1.2% gel after confirming the bands. This time, 20 μL of DNA loading dye was added to each sample and 65 μL of mixture was injected into the gel. This gel ran for 40 minutes at 90 V before imaged. Next, razors were used to extract each band from the gel using UV exposure to visualize the bands. Each extraction was inserted into a labeled Eppendorf tube. To complete this step, the Thermo Scientific gel extraction kit was followed, and the product was stored in the -20 ℃ freezer. In 2nd PCR, one PCR tube is labeled "2nd PCR Product" and the following reagents are added: 58 μL of nH_2O, 20 μL of 5x Phusion Buffer, 5.0 μL of DMSO, 2.0 μL of 10 mM dNTPs, 2.0 μL of BamH1 Forward Primer, 2.0 μL of EcoRV Reverse Primer, 10 μL of Forward Mutant Fragment, 10 μL of Reverse Mutant Fragment, and lastly 1.0 μL of Phusion Plus DNA polymerase. This mixture was run on the thermocycle and checked on a 1% gel. The Thermo Scientific PCR purification kit was then used to purify this PCR product. Restriction Digestion was then performed by adding the following reagents in an Eppendorf tube labeled "Mutant Insert:" 50 μL of purified 2nd PCR, 2.0 μL of BamHI-HF Enzyme, 2.0 μL of EcoRV-HF Enzyme, 6.0 μL of 10x Cutsmart Buffer. The following was also added to an Eppendorf tube labeled "Vector:" 1 μg of Vector stock, 1.0 μL of BamHI-HF Enzyme, 1.0 μL of EcoRV-HF Enzyme, and 5.0 μL of 10x Cutsmart Buffer. Both of these Eppendorf tubes were incubated at 37 ℃ for 1 hour. To stop the reaction of the Mutant insert, 20 μL of DNA loading dye was resuspended into the respective tube and stored in the -20 ℃ freezer. The Vector incubated for an additional hour after 2.0 μL of CIP was added. This reaction was stopped identical to the Mutant insert and stored. Gel extraction was then performed using the Thermo Scientific Gel Extraction kit protocol. A recombinant plasmid was then generated through ligation by combining the following in a tube: 8.0 μL of nH_2O, 1.5 μL of 10x Ligase Buffer, 1.5 μL of Vector, 3.0 μL of Mutant Insert, and 1.0 μL of T4 DNA Ligase. These reagents were combined and set at room temperature for 20 minutes before being stored in the -20 ̊C freezer. Next, a transformation was performed by combining the following: 73.0 μL of nH_2O, 20.0 μL of 5x KCM Buffer, 7.0 μL of Recombinant Plasmid, and 100 μL of E. coli competent cells (XL 1 Blue for pcDNA 3.1+). This tube cooled on ice for 20 minutes before 500 μL of LB Broth was added using the aseptic technique. This product was then run on the shaker at 250 rpm and 37 ̊C for 40 minutes. The tube was removed and ran on a tabletop centrifuge for 2 minutes at 2000 rpm. Following this, 500 μL of supernatant was discarded as waste and the pellet is resuspended into the remaining volume. Using the aseptic technique, 200 μL was spread on an LB Agar ampicillin plate and spread with 5 autoclaved glass beads. This plate was incubated for 16 hours at 37 ̊C. When completing Mini Preps Purification, 2 mL of LB and 2 μL of ampicillin were added to 3 round bottom falcon tubes using the aseptic technique. Three separate pipette tips were used to select one colony from the plate and inserted into a tube. The plate was placed on the shaker at 250 rpm and 37 ̊C for 16 hours. Next, the Mini Prep Purification Kit was followed and the remaining 500 μL of product was kept for a future step. For Mini Prep Screening, the following reagents were mixed in a tube: 7.5 μL of nH_2O, 5.0 μL of Mini Prep, 1.0 μL of BamH1-HF, 1.0 μL of EcoRV-HF, and 1.5 μL of 10x r-Cutsmart Buffer. This tube was incubated for 1 hour and 8 μL of DNA loading dye was resuspended to stop the reaction. These three products were checked on a 1% agarose gel and the concentrations were checked on a nanodrop to determine which product will be sent for Sanger Sequencing.

Results

When making an acetylated lysine residue, we began with first polymerase chain reaction (PCR). In this experiment we run a two-step PCR because it is critical to create the mutation during 1st PCR while generating the forward and reverse fragment, and then combining these during 2nd PCR. If only one PCR was completed, the mutations would occur at the end of the sequence instead of in the middle. These mutations must be in the middle to properly assess splicing activity. This mutation must be introduced to determine if post-translational acetylation affects the function of RNA binding proteins.

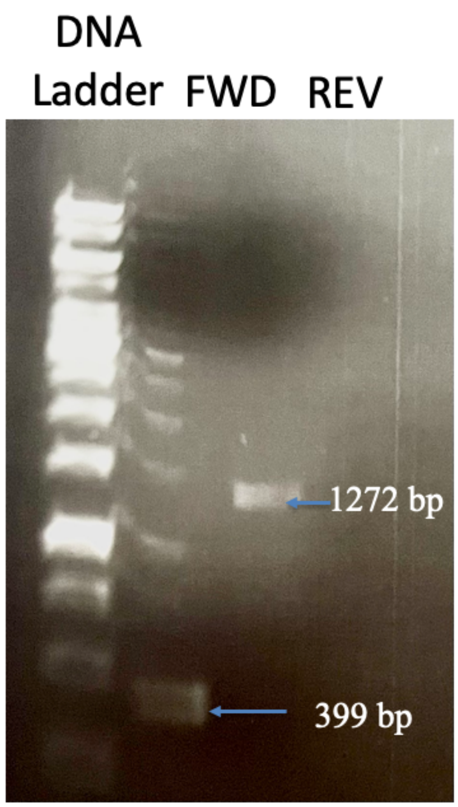


Figure 1: This image illustrates 1st PCR, with the K424 forward fragment on the left and the reverse fragment on the right on a 1% agarose gel.

The 1st PCR products were made with reagents listed in the methods section and confirmed by running on a gel with electrophoresis. The forward fragment is located at 399 base pairs and the reverse fragment is located at 1272 base pairs. Based on these two bands being in an accurate location according to the DNA ladder on the far-left lane, the same samples were run on a 1.2% gel for gel extraction.

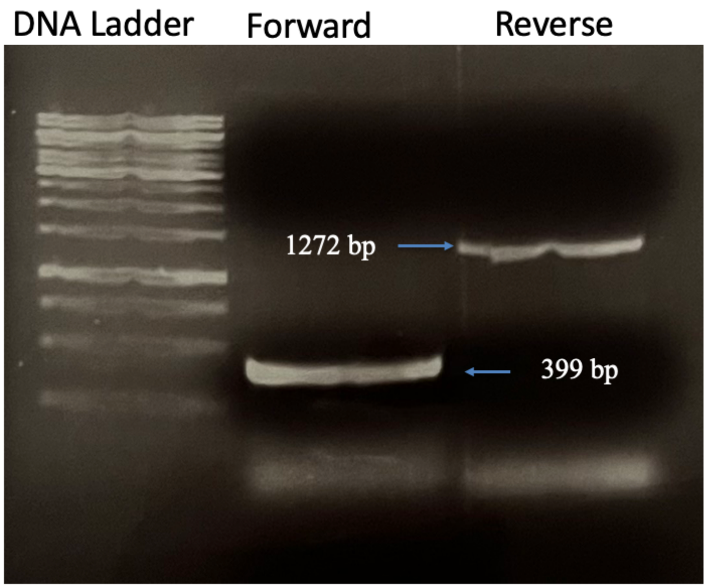


Figure 3: This image is the forward and reverse fragments from 1st PCR but run on a 1.2% gel for extraction. The DNA ladder is located on the far left, followed by the forward, and then the reverse.

To use the forward and reverse 1st PCR products for the 2nd PCR, gel extraction was completed. These two bands were extracted with a razor blade and the Thermo Scientific Gel Extraction kit protocol was followed to prepare for 2nd PCR.

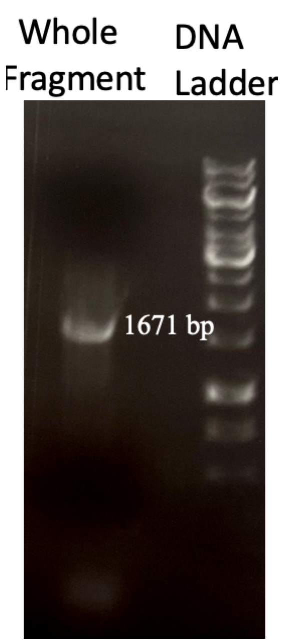


Figure 3: This reveals the 2nd PCR product located at 1671 base pairs, which is also the full length K424 mutant.

Note that adding the size of the forward and reverse fragment generated in figure 1 equals the size of the band in the 2nd PCR product. This band makes up the forward and reverse fragment that have now been combined into one. Now that this band size was confirmed, restriction digest was used to generate a mutant insert and a vector.

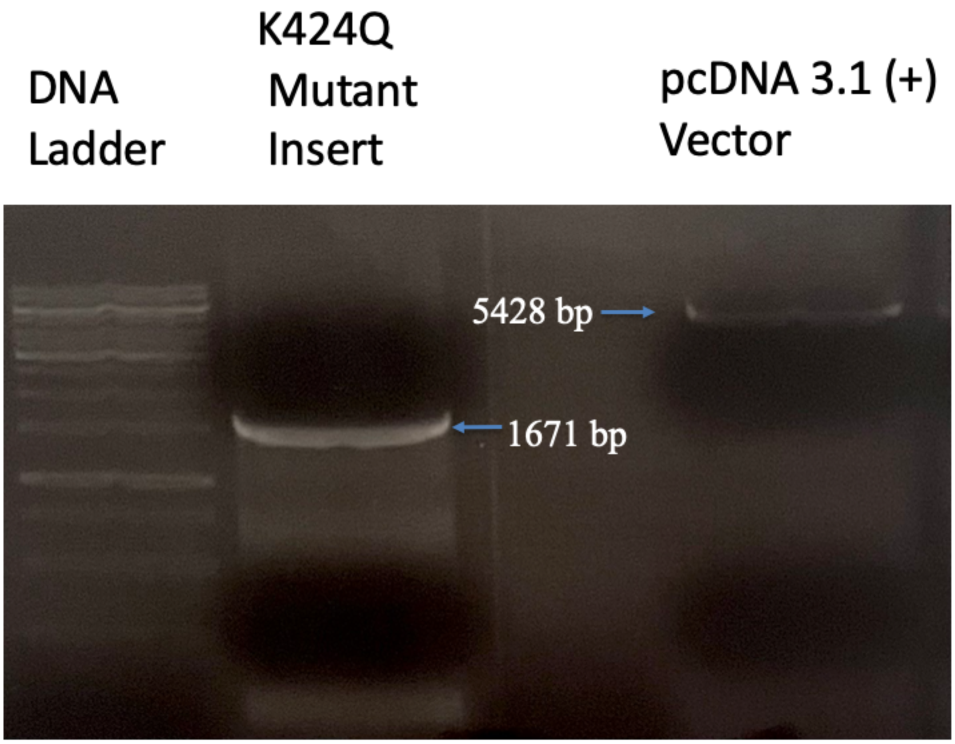


Figure 4: This image shows the K424 mutant insert on the left and the vector on the right on a 1.2% gel. The DNA ladder is on the left, the mutant insert is in the middle at 1671 base pairs, and the vector is on the right at 5428 base pairs.

During restriction digest, two different enzymes were used to cut open and make sticky ends of the vector and mutant insert. These sticky ends are covalently bonded during ligation to form a recombinant plasmid. This recombinant plasmid is spread on a transformation plate to grow colonies.

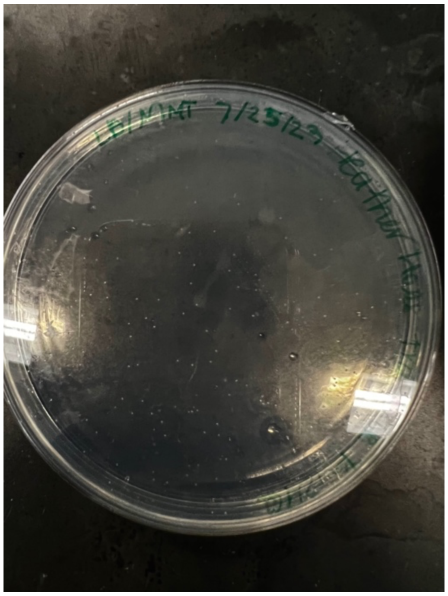


Figure 5: This image is of the completed transformation with colonies. The recombinant plasmid was grown in competent cells on LB/Amp agar plate at 37 degrees Celsius for 16 hours.

Three of the resulting colonies were selected and Mini Prep was completed. During this step, the colonies were grown in LB/Amp test tubes at 37 degrees Celsius for 16 hours.

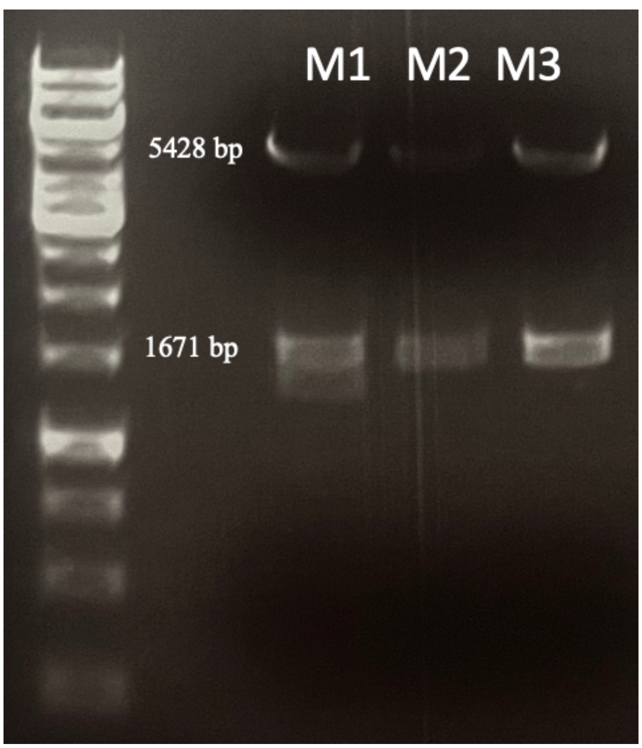


Figure 6: This illustrates the 3 mini preps prepared. All 3 Mini Preps were successful with the vector at 5428 base pairs and the mutant insert at 1671 base pairs.

Each of these Mini Preps had bands that appeared at the correct location for the K424Q mutant and were each of these were sent for Sanger Sequencing. None of these Mini Preps provided a successful forward and reverse sequence, so the entire methods were repeated. The results were identical to figures 1-5, ending at the transformation plate. Three colonies were again selected and grown in LB/Amp test tubes.

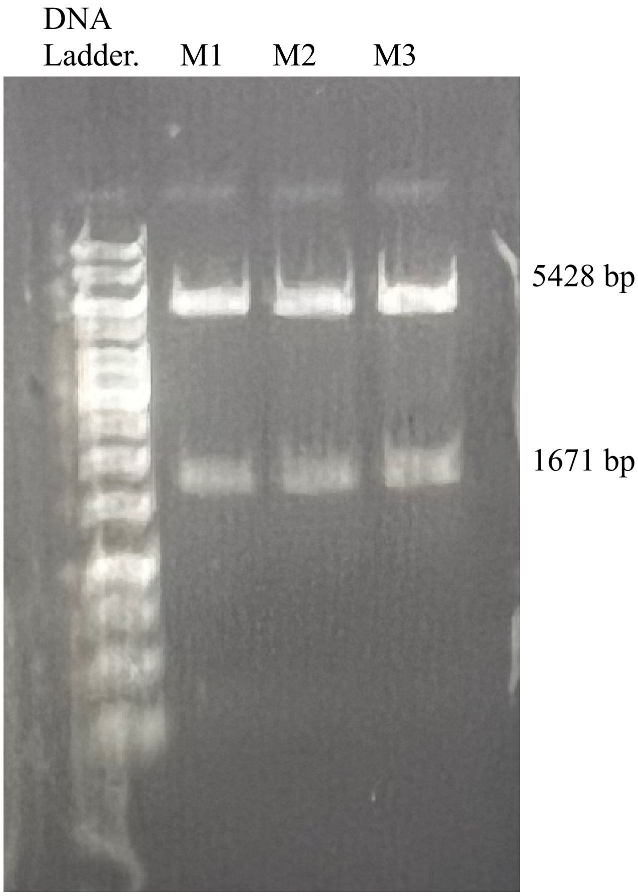


Figure 7: This illustrates the 3 mini preps prepared. All 3 Mini Preps of trial 2 were successful with the vector at 5428 base pairs and the mutant insert at 1671 base pairs.

These new Mini Prep samples revealed bands at the correct location for the K424Q mutant and were each sent for Sanger Sequencing to confirm or deny the sequence. The concentrations were as follows: M1 was 64.5 ng/L, M2 was 90.7 ng/L, and M3 was 153.2 ng/L.

Discussion

The objective of these methods was to combine fragments to create a full length PTBP1 mutant, K424Q. The 1st PCR in figure 1 was successful with the expected 399 base pairs for the forward fragment and the 1272 base pairs for the reverse. In lane 2, there were additional bands due to the DNA ladder seeping over from lane one. This was a pipetting error because the DNA ladder was not injected directly into lane 1 and the tip was pulled out before it was empty. However, this 1% gel was just to confirm the band size of the forward and reverse, so figure 2 displays a more accurate reading of the 1st PCR product. Moving on to figure 3, the 2nd PCR product appeared at 1671 base pairs, which was an expected location for the combination of the forward and reverse product of 1st PCR. Note that 399 base pairs for the forward fragment added to the 1272 base pairs for the reverse fragment equals the full length PTBP1 mutant, specifically K424Q. Figure 3 does display slight streaking, likely due to not changing the 1% TAE buffer in the gel electrophoresis. This mistake can lead to poor separation due to depletion of ions and SDS in the buffer. Restriction digest appeared successful in figure 4 with the mutant insert at 1671 base pairs and the vector at 5428 base pairs. There was minimal streaking for the mutant insert during this step and the bands were bright and defined at the correct location. Figure 5 reveals that the transformation plate was successful, with an abundance of colonies that were isolated. It is ideal to choose colonies that are isolated because it allows for consistency and accuracy and ensures that the colony is not contaminated with additional bacteria. After completing the Mini Prep in figure 6, each colony was successful in producing the mutant insert at 1671 base pairs and the vector at 5428 base pairs.

Each of these Mini Preps were sent for Sanger sequencing and the reverse sequence was successful for each, but the forward sequences were missing the FLAG tag. A FLAG tag is the amino acid sequence DYKDDDDK, that is necessary for future research. It is likely that this

FLAG tag was cut out with the blade during gel extraction and not included when forming the 2nd PCR product. Due to the K424Q sequence not being successfully confirmed, it was essential to restart the methods from 1st PCR. Going back to the first step would result in new products and the chance for a successful PTBP1 mutant to be created.

In the second trial, the same steps were followed as in trial 1. The 1st PCR, 2nd PCR, and Restriction Digest products appeared at the same locations as in the first trial. Also, the transformation plate grew an appropriate number of colonies. Similar to trial 1, the 2nd trial gave 3 successful Mini Prep results that were all sent for sequencing. The concentrations of the mini preps in trial 2 were 64.5 ng/ uL, 90.7 ng/ uL, and 153.2 ng/uL. It is recommended that the concentrations are above 80 ng/uL, so product 3 was sent in for sequencing first. The reverse sequence was confirmed, but the forward sequence was missing the first 10 Amino Acids and the FLAG tag. This is likely due to the primer being old because this primer is from June of 2023. The concentration of product 2 is still above 80 ng/uL, so that product was also sent for sequencing. This sequence is still waiting for conformation and if it comes back successful, a Midi Prep will be performed, which is a large-scale Mini Prep. This plasmid will later be used in the lab to assay the mutant for splicing activity and answer the hypothesis.

References

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